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Contact Monitor:		**Contact Phone:**	
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COI Name (Institution):	Heldt, Thomas (Massachusetts Institute of Technology) Verghese, George (Massachusetts Institute of Technology)		
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	One of the highest priority problems in the current manned space program is orthostatic intolerance (OI) experienced by astronauts upon their return to the normal gravitational environment. This problem has been well known since the earliest days of manned space flight, and has been intensely studied in-flight and in many ground-based (bed-rest) studies. A number of countermeasures have been proposed and evaluated, but to date no effective and practical countermeasure has been developed. While the effectiveness of countermeasures can only be explored in ground-based experiments and ultimately be proven successful when used by astronauts, the small number of subjects studied in both situations makes the interpretation of limited experimental data for countermeasure development and refinement difficult. Mathematical models that represent the essential functional components of the cardiovascular system provide a powerful		

Task Description:	means to extract physiologically relevant information from multiparameter experimental data. We will extend progress already made, with the following specific aims: 1. Develop procedures and algorithms for automated model-based extraction of physiologically relevant information from multivariate data streams, particularly those relevant to countermeasure development for OI. We will expand our unified data archive of hemodynamic signals during gravitational stress. 2. Develop strategies for systematic and effective model identification, i.e., for the adaptation of model structures and parameters to the characteristics of the available physiological data. These strategies will be based on analytical and computational studies of candidate models and their sensitivities to parameter changes. 3. Apply our computational modeling and parameter identification technology to the challenge of cardiovascular monitoring of critically ill patients. Hemodynamic measurements will be transformed into physiologically meaningful parameters through model-based processing. 4. Develop and thoroughly test a graphical user interface for our current cardiovascular model. The resultant cardiovascular simulator will be open-source and capable of running on all major operating systems. Furthermore, the entire model will be made publicly available to researchers and educators. Major progress was made during the first 12 months of the program: - Our simulations suggest that the differences between the hemodynamic responses to passive head-up tilt and to active standing can be attributed primarily to mechanically-mediated rather than autonomically-mediated events. - Simulation of a range of current hypotheses of the mechanisms of orthostatic intolerance revealed that total blood volume has by far the greatest impact on blood pressure homeostasis in the upright posture. If the volume status is euvolemic, large alterations in other parameters (such as arterial resistance or venous tone feedback, cardiac parameters, or leg venous compliance) must be hypothesized before blood pressure homeostasis becomes impaired. However, if the volume status is even borderline hypovolemic, relatively small changes in other parameters can have significant effects on blood pressure homeostasis in the upright posture. - Our simulations indicate that the gain of the venous tone reflex is among the most important secondary parameters responsible for blood pressure homeostasis in the upright posture. This result has implications for countermeasure development as it suggests that an alpha-adrenergic agonist, such as midodrine, can help alleviate the drop in blood pressure that is associtated with post-spaceflight orthostatic intolerance. - We demonstrated the ability of our parameter identification algorithm to recover with good fidelity parameter values from simulated, noise-corrupted hemodynamic data streams. When applied to astronaut data pre- and post-spaceflight, the estimation algorithm identified a spaceflight-related reduction in blood volume of 9% --- a value well within the range of reported measurements. - We have adapted the model for the context of parameter estimation from critically ill patients in intensive care units (ICUs). Initial attention focussed on detection and tracking of hemorrhage. Early results demonstrate that we are able to track the relative change in blood volume from simulated data. The results obtained to date demonstrate that the models developed provide a rational framework with which to interpret hemodynamic data and allow for comparison of the relative impact deleterious changes in cardiovascular parameters can have on overall performance of the cardiovascular system. The research presented in this progress report is well within the time frame set forth in the original proposal; the progress made so far does not necessiate any changes to the specific aims of the proposal or the strategy with which the research is pursued. We intend to pursue the following goals during year 2 of the funding period: - Apply the parameter estimation algorithm to more astronaut data. As the Shuttle is returned to service, we hope to continue our productive collaboration with the Cardiovascular Laboratory at Johnson Space Center and offer our computational models to interpret the hemodynamic data obtained from astronauts before and after space travel. - Develop further model reduction and identification methodologies that allow for robust and efficient parameter estimation from a variety of hemodynamic data streams. - Begin development and testing of the platform-independent graphical user interface to make available our computational models to the wider scientific community. Although outside the scope of the currently funded project, but certainly within the spirit of the program, we have recently joined forces with members of the Man Vehicle Lab at MIT to use our model for the interpretation of hemodynamic data from subjects undergoing short-radius centrifugation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Outside the scope of modeling the effects of microgravity, numerical cardiovascular simulations 1. provide a basis for interpretation of clinical data in the context of intensive care. Simulations are basic components of an intelligent patient monitoring system which can couple real-time physiologic data to cardiovascular pathology. 2. provide a powerful environment for teaching of normal and abnormal cardiovascular physiology.
	Progress during the first year of funding proceeded in the following topical areas: 1) Model refinement and exploration, 2) Model analysis and estimation, and 3) Model extension. 1) Model refinement and exploration: We have succeeded in simulating the hemodynamic responses to passive head-up tilt and active standing-up within the framework of a single computational model (a feat that has eluded modelers so far). Our simulations call into question the prevailing explanation of the differences between the hemodynamic responses to passive head-up tilt and actively standing-up; the models generate testable hypotheses and therefore suggest experiments to validate our simulations. All parameters of the cardiovascular system scale with body size (total blood volume has a particularly strong

Task Progress:	dependence on body weight, for example). For simulation of different-sized individuals, we have devised a scaling scheme that allows for scaling of cardiovascular parameters based on body weight. Such scaling captures the effects of size on cardiovascular performance and allows for population simulations rather than the simulation of a "representative 70-kg male subject". 2) Model analysis and estimation: Increasing the complexity of a model allows for better representation of cardiovascular physiology on a variety of time and length scales. However, increasing the complexity of a model can also work significantly against its usefulness in many respects: parameter values become harder to pick and it becomes more difficult to understand on a fundamental level which parts of the model are actually being exercised and how. To balance this trade-off between model complexity and simplicity, we have continued to conduct extensive sensitivity analyses on our models in order to determine which parameters significantly impact a particular model output. This knowledge is in turn critical to matching a model to measured data by parameter tuning. To this end, we have conducted first- and second-order local sensitivity analyses on the full model; we explored the possibility of using reduced-order models; and we have implemented and tested parameter estimation algorithms. Of particular note is the sensitivity of mean arterial pressure to changes in parameters that have been implicated in the genesis post-spaceflight orthostatic intolerance: venous tone and arterial resistance reflex gains, right ventricular end-diastolic compliance, reductions in total blood volume, and increases venous compliance of the legs. 3) Model extension: We have begun to extend the scope of our models to the problem of clinical decision support for applications both in space and on Earth. To this end, we have focused out attention on the problem of patient state estimation from hemodynamic data streams (such as ECG, arterial blood pressure, and central venous pressure) that are common in modern intensive care units. Our initial effort revolved around the identification and the
Bibliography Type:	Description: (Last Updated: 08/08/2019)
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Presentation	Heldt, T and Renijfo, C. and Samar, C. and Verghese, G. C. "Assessing Cardiovascular Function using Electrical Circuit Analogs " Feb-2005
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Presentation	Heldt, T. and Mark, R. G. "Computational Models of Cardiovascular Response to Orthostatic Stress " Oct-2004
Presentation	Heldt, T. and Mark, R. G. "Modeling of Cardiovascular Response to Orthostatic Stress " Oct-2004
Presentation	Heldt, T. and Mark, R. G. "Scaling Cardiovascular Parameters for Population Simulations " Sep-2004